Contribution Record

Project: E-Commerce & Order Management System (Olist) · **Group:** TarushTanishkTanishq
Course: CSN15101 · Software Engineering and Project Management · **Instructor:** Komal Gupta

1. Team

Member	Registration no.	Role
Tarush Banke	20243293	Project lead, Scrum master · database architect
Tanishk Varshney	20243288	Back-end and database-logic developer
Tanishq Mahajan	20243289	Front-end and ML developer · QA / test lead

2. Task-allocation table

WBS	Task	Owner	Support	Status
1.1	Planning, sprint board, risk register, meeting minutes	Tarush	–	Done
1.2	Dataset profiling; SRS	Tarush	Tanishq	Done
1.3.1–1.3.2	ER/EER model, relational schema, keys, FDs, BCNF	Tarush	–	Done
1.3.3–1.3.4	Staging load, cleaning, transform; indexes and views	Tarush	–	Done
1.3.5	Triggers and stored procedures	Tanishk	–	Done
1.3.6	Security: roles, grants, column privileges, RLS	Tanishk	–	Done
1.3.7	Transactions and concurrency demos	Tanishk	–	Done
1.4.1–1.4.2	DB access layer, Flask API, report queries Q1–Q16	Tanishk	–	Done
1.4.3–1.4.4	Front end; ER model & schema page	Tanishq	–	Done
1.5	ML: framing, feature view, training, evaluation, integration	Tanishq	Tarush (feature view), Tanishk (API)	Done
1.6.1	Test plan and test execution	Tanishq	All	Done
1.6.2	Design document, README	Tarush	–	Done
1.1.3	Mid-semester report and demo video	All	–	Done

3. Individual contributions

Member	Main contributions	Main files	Commits	Lines added*
Tarush Banke	Project planning; SRS; ER/EER model; schema; keys, FDs, BCNF; CSV staging load and cleaning; indexes; views and materialized views; DDL/DML, index and normalization demos; design doc, project plan, README	`db/00` – `05`, `db/12` – `14`, `setup_db.bat`, `docs/SRS.md`, `docs/Design.md`, `docs/ProjectPlan.md`, `docs/er_diagram.png`	25	2,241

Member	Main contributions	Main files	Commits	Lines added*
Tanishk Varshney	Triggers; stored procedures and functions; roles, grants, column privileges, RLS; 16 report queries; ACID and concurrency demos; Flask API and DB access layer	`db/06 – 11`, `backend/app.py`, `backend/db.py`, `backend/queries.py`, `run_app.bat`	24	2,573
Tanishq Mahajan	Front end (all pages, SQL panel, faceted search, lab, ER page); ML framing, features, model selection, training and evaluation; ML service; test plan, testing and screenshots	`frontend/`, `ml/`, `backend/ml_service.py`, `train_model.bat`, `docs/TestPlan.md`, `docs/screenshots/`	18	4,964

**Lines added from `git log --numstat`. Tanishq's count includes the bundled third-party `frontend/vendor/mermaid.min.js`, so it overstates hand-written code.*

All members took part in sprint planning, reviews and testing, and each can explain the complete flow Front end → SQL → Database → Result.

4. Git commit history (evidence)

Repository: `github.com/TarushB/SEPM-Ecommerce-Order-Management`, branch `main`, 67 commits from 18 Sep to 29 Sep 2026, each authored by the member who did the work.

Commit	Date	Author	Message
`2fdfa6e`	2026-09-18	Tarush Banke	Initial commit: project skeleton, gitignore and README
`3297e5c`	2026-09-18	Tanishk Varshney	Add Python requirements (Flask, psycopg, pandas, scikit-learn)
`52e514f`	2026-09-18	Tarush Banke	Schema: state, zip_code and customer tables
`67956be`	2026-09-18	Tarush Banke	Schema: seller, category and product with CHECK constraints
`051ae9d`	2026-09-18	Tarush Banke	Draft project plan: process model, roles, WBS
`54939ee`	2026-09-19	Tarush Banke	Schema: orders, order_item, payment and review
`36e9f09`	2026-09-19	Tarush Banke	Add SRS draft: user classes and functional requirements
`9b798ab`	2026-09-19	Tarush Banke	Schema: audit log, ML tables and app_user
`accaacc`	2026-09-19	Tarush Banke	Add staging schema and CSV bulk load
`79d9a6b`	2026-09-19	Tanishq Mahajan	Test plan outline: objectives, scope, approach
`511688c`	2026-09-19	Tarush Banke	SRS: non-functional requirements, use cases, traceability
`af5a4c4`	2026-09-20	Tarush Banke	Transform: states and one zip row per prefix from geolocation

Commit	Date	Author	Message
b2823fd	2026-09-20	Tarush Banke	Transform: customers, sellers, categories and products
2c73010	2026-09-20	Tarush Banke	Transform: orders, items, payments and reviews
5e836c0	2026-09-20	Tarush Banke	Add ER diagram
d2f4250	2026-09-21	Tanishk Varshney	Trigger: order status can only move forward
40164a2	2026-09-21	Tarush Banke	Add B-tree, composite, partial and GIN full-text indexes
9f82613	2026-09-21	Tanishk Varshney	Trigger: audit trail for order status changes
e6b6351	2026-09-21	Tarush Banke	Add run-all script and Windows setup_db.bat
8c4cb1c	2026-09-21	Tanishk Varshney	Triggers: review only after delivery, block deleting delivered orders
09b773b	2026-09-22	Tarush Banke	Views: order details, monthly sales, seller performance
b9c6559	2026-09-22	Tanishk Varshney	Triggers: invalidate ML prediction, stock control
1e8d3d0	2026-09-22	Tanishk Varshney	Procedure sp_place_order: order, items and payment in one transaction
f3a7a34	2026-09-22	Tarush Banke	Views: updatable view WITH CHECK OPTION, data-quality view
22ef1d3	2026-09-22	Tanishk Varshney	Procedures sp_update_status and sp_cancel_order
c2a7cbe	2026-09-23	Tarush Banke	Materialized feature views for the ML model
9ac58d3	2026-09-23	Tanishk Varshney	Functions fn_login (bcrypt), fn_order_total, fn_seller_rating
1ab7273	2026-09-23	Tanishq Mahajan	ML: shared feature list and database I/O helpers
fc3cdb0	2026-09-23	Tanishk Varshney	Security: group roles, least-privilege app login, grants
5e2d100	2026-09-23	Tanishq Mahajan	ML: load features from SQL, time-series cross-validation
972a2cd	2026-09-23	Tanishk Varshney	Security: column privileges, row-level security, demo users
213b4dd	2026-09-24	Tanishq Mahajan	ML: compare baseline, logistic regression, random forest, HGB
394a083	2026-09-24	Tanishk Varshney	Report queries Q1-Q8: joins, subqueries, GROUP BY/HAVING
407d85a	2026-09-24	Tanishq Mahajan	ML: threshold tuning, test on Jun-Aug 2018, store results in DB

Commit	Date	Author	Message
d387790	2026-09-24	Tanishk Varshney	Report queries Q9-Q16: windows, CTEs, set ops, full-text
ef49f9a	2026-09-24	Tanishq Mahajan	Add train_model.bat
eff59ec	2026-09-24	Tanishk Varshney	Export report queries to db/09_queries.sql
140a47a	2026-09-25	Tanishk Varshney	DB layer: per-request transaction, SET LOCAL ROLE, SQL log
6f65431	2026-09-25	Tanishq Mahajan	ML service: load model bundle and score orders
ed479f1	2026-09-25	Tanishk Varshney	API: app setup, helpers, login and dashboard
4e707ad	2026-09-25	Tarush Banke	README: setup steps, database design, SQL coverage
f964b75	2026-09-25	Tanishk Varshney	API: order search with facet counts, order detail, place/update/cancel
5de50c2	2026-09-26	Tanishq Mahajan	Front end: page shell and base styles
ae8a549	2026-09-26	Tanishk Varshney	API: CRUD for products, customers, sellers, reviews, categories
3970292	2026-09-26	Tanishq Mahajan	Front end: API helpers, login screen, SQL panel
97e0a31	2026-09-26	Tanishk Varshney	API: reports with EXPLAIN and CSV, read-only SQL console, schema
137db1e	2026-09-26	Tanishk Varshney	Add run_app.bat
93bfc3b	2026-09-26	Tanishq Mahajan	Front end: dashboard and faceted order search
edc5c17	2026-09-27	Tanishk Varshney	API: ML summary/predict, transactions lab, user admin
f26a5cd	2026-09-27	Tanishq Mahajan	Front end: order detail, place order, CRUD pages
3a7b3c5	2026-09-27	Tanishq Mahajan	Front end: reports page and SQL console
2b751a6	2026-09-27	Tanishq Mahajan	Front end: product, customer and seller forms
b46dbf6	2026-09-27	Tanishq Mahajan	ER model & schema page with bundled Mermaid
1cf812c	2026-09-27	Tanishq Mahajan	Front end: transactions lab, ML insights, user admin
b6e4f5b	2026-09-28	Tanishk Varshney	Add ACID transactions demo script
499bd14	2026-09-28	Tarush Banke	Add DDL/DML demo script

Commit	Date	Author	Message
`8f85358`	2026-09-28	Tanishk Varshney	Add concurrency demo: lost update, isolation levels, deadlock
`f38f792`	2026-09-28	Tarush Banke	Add EXPLAIN ANALYZE index demo
`72c8ac3`	2026-09-28	Tanishq Mahajan	Test evidence: dashboard, order search, order detail, reports
`01891f0`	2026-09-28	Tarush Banke	Add normalization demo: anomalies and FD checks
`8c2ba60`	2026-09-28	Tanishq Mahajan	Add test plan with test cases TC-01..TC-31
`97c885f`	2026-09-28	Tarush Banke	Add system design document
`6cccbff`	2026-09-29	Tanishq Mahajan	Test evidence: transactions lab, ML, seller RLS, ER page
`59906e7`	2026-09-29	Tanishk Varshney	Fix: serve index.html at / (was 404)
`de64c36`	2026-09-29	Tanishk Varshney	run_app.bat: check PostgreSQL is installed and running first
`7524bd3`	2026-09-29	Tarush Banke	Project plan: schedule, COCOMO estimate, risk register, minutes
`427aeb6`	2026-09-29	Tarush Banke	Rewrite README for SEPM: process model, requirements, testing, contributions

5. Use of generative AI tools (disclosure)

Tool	Purpose	Nature of generated content	Verification / modification by students
Claude (Anthropic)	Assistance with parts of the code during development; writing the automated test scripts (`db/15_test_suite.sql`, `tests/api_tests.py`); creating the project report and submission documents	Code suggestions, test scripts, report text, diagrams and formatting	All members reviewed, tested and modified the content; the team ran the system and can explain every part

6. Individual contribution declaration

We confirm that the contributions recorded above are accurate, that the submitted work was prepared by our group, and that any use of AI tools is disclosed in section 5.

Name	Registration no.	Signature / confirmation	Date
Tarush Banke	20243293	Confirmed	15 Oct 2026
Tanishk Varshney	20243288	Confirmed	15 Oct 2026
Tanishq Mahajan	20243289	Confirmed	15 Oct 2026